# Vidya Sagar Tirandas

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.52 |
| Narayana High School | Class XII (CENTRAL BOARD OF SECONDARY EDUCATION) | 2021 | 87.00% |
| De Paul School | Class X (INDIAN CERTIFICATE OF SECONDARY EDUCATION) | 2019 | 89.80% |

## SKILLS

**Programming & Data:** Python, SQL, Pandas, NumPy, Scikit-learn

**Data Analysis & Visualization:** EDA, Data Cleaning, Statistical Analysis, Regression, Matplotlib, Seaborn, Tableau

**Machine Learning:** Classification, KNN, Logistic Regression, Clustering, Model Evaluation

**Deep Learning & Computer Vision:** TensorFlow, Keras, CNNs, YOLO, OpenCV, MediaPipe, Object Detection, Image Classification

**NLP & Data Collection:** NLP, Tokenization, Text Classification, BeautifulSoup, Selenium

**Tools & Platforms:** Jupyter Notebook, Google Colab, Git, VS Code

## PROJECTS

### Real-Time Gesture-Based System Volume Controller

- Developed a **real-time computer vision interface** using OpenCV and MediaPipe for contactless system-volume control through hand gestures.
- Enabled **gesture-based volume adjustment, mute/unmute, and control lock/unlock** using real-time hand landmark tracking.
- Achieved **real-time operation at 17-22 FPS** with live visual feedback for volume level, gesture distance, and control status.

### Custom Object Detection using YOLOv8

- Built a custom dataset of **4,127 images across 13 project-specific classes** using publicly available images and Roboflow.
- Trained a **YOLOv8n** object detection model achieving **82.4% mAP@50, 82.9% precision, 76.8% recall, and 79.7% F1-score**.
- **Exported the trained model to ONNX** and deployed it for **CPU-based real-time inference**, achieving ~**14 FPS** with ~**1 GB RAM** usage.

### LLM-Driven Natural-Language Control of a ROS 2 Robot

- Built an **LLM-driven task-planning pipeline** that translates natural-language instructions into structured plans that Nav2 carries out on a TurtleBot3 in Gazebo. The LLM backend can be switched between local (Ollama) and cloud (Gemini) with one launch argument.
- Implemented a **safety-constrained execution framework** with **8 permitted actions**, target and parameter validation, and automated plan repair, achieving **100%** correct plans on a **26-command** benchmark, and **54%** with a regex baseline.
- Integrated **camera-LiDAR-based object localization** with an error of 0.03-0.24 m for improved perception and

navigation.

## THESIS

**Multi-Robot Exploration and Collaboration System**

- Developed a **decentralized 3-robot system** using ROS2 and Gazebo for unknown-maze exploration, target detection using camera-LiDAR fusion, and cooperative transportation of a box requiring multiple robots.
- Implemented a custom autonomy stack without Nav2, featuring **occupancy-grid mapping**, **inter-robot map sharing**, coordinated frontier-based exploration, **A* path planning**, and pure-pursuit control.
- Designed **auction-based task allocation and cooperative pushing**, with sequential help requests, synchronized robot coordination, and push-face switching at corners to navigate the box through constrained spaces.

## INTERNSHIP

**Research Intern, DTE 4 Calamity and Humanity Pvt Ltd.**

- Developed a **ROS2-based real-time perception pipeline** integrating camera, LiDAR, and mmWave radar for human detection and localization.
- Implemented **YOLO-based object detection, field-of-view filtering, and radar-camera target association**(**5° angular association tolerance**) for multi-person perception.
- Processed **640 × 480 camera input at 30 FPS** and designed radar-based target association for up to **3 human targets**, using modular ROS 2 nodes and **approximate time synchronization and TF transformations**, with real-time deployment and testing on Raspberry Pi.

## POSITIONS OF RESPONSIBILITY

**Placement Training Examination Volunteer | IIT Hyderabad**

- Prepared practice questions and assessment material to help students from the Department of Mechanical & Aerospace Engineering prepare for online placement assessments.

**Campus Ambassador | VISIT IITH club**

- Represented IIT Hyderabad by engaging with and facilitating visits for school students, international delegates and industry professionals with showcasing institute's academic and research ecosystem.

**Social Media & Content Coordinator | Akshara Sanjeevani club**

- Manage Club's social presence and contribute to content strategy, creative ideation and event initiatives.

## RELEVANT COURSES

**Relevant courses:** Data analytics, Machine Learning and its Applications, DBMS, Autonomous Robotics Systems, Fundamentals of Robotics, Control System Design, Computer Vision, Sensors and Actuators, Soft Robotics, Probability and Optimization.